Annexure A
CO2 AT3

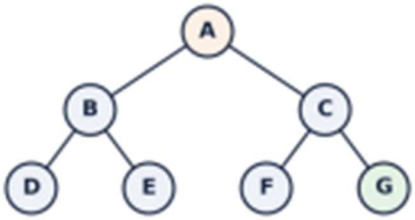
Dry Run Evaluation

CSA17 Artificial Intelligence · Searching Techniques ·Slot D (CSA1711)

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Instructions: trace each algorithm by hand. Fill in every table and answer the questions below it. Show each step; a final answer without a trace gets no marks.

Question 2 – Depth-First Search (DFS)



Task: Using the same graph, perform Depth-First Search (DFS) starting from A, with goal G. Push children right to left so the left child is explored first. Complete the table (top of the stack is on the right).

Iteration	Stack Before (top at right)	Node Visited	Stack After
1	[A]	A	[C, B]
2	[C, B]	B	[C, E, D]
3	[C, E, D]	D	[C, E]
4	[C, E]	E	[C]
5	[C]	C	[G, F]
6	[G, F]	F	[G]
7	[G]	G	[]

Questions:

1. Write the DFS traversal order.
The tree is:
A

/ \
B C
/ \ / \
D E F G

DFS explores one branch completely before moving to the next branch.

Traversal order:

A → B → D → E → C → F → G

2. How many nodes are visited before reaching G?

Answer:

6 nodes are visited before reaching G. Including G, the total number of visited nodes is **7**.

3. Compare the memory needed by BFS and DFS on this tree.

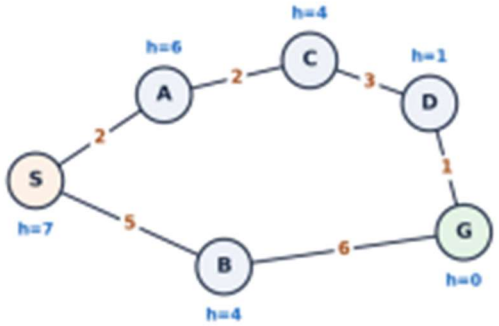
ANS:

Compare the Memory Needed by BFS and DFS

- **BFS:** Generally requires more memory because it stores many nodes at the same level.
- **DFS:** Generally requires less memory because it stores the current path and unexplored branches

CSA17 Artificial Intelligence · Annexure A · Dry Run Evaluation

Question 7 – Greedy Best-First Search



Task: Find a path from S to G using Greedy Best-First Search, f(n) = h(n). Edge numbers are step costs; h values are written next to each node. Stop when G is selected.

Iteration	Frontier Before (Node, h)	Node Expanded	Frontier After
1	[(S, 7)]	S	[(B, 4), (A, 6)]
2	[(B, 4), (A, 6)]	B	[(G, 0), (A, 6)]
3	[(G, 0), (A, 6)]	G	Goal reached

Questions:

1. Write the path found and its total cost.

The path selected is:

$$S \rightarrow B \rightarrow G$$

Edge costs:

- S → B = 5
- B → G = 6

Therefore,

$$\text{Total Cost} = 5 + 6 = \boxed{11}$$

Path = S → B → G; Total cost = 11

2. Is this path optimal? Why or why not?
No, the path is not optimal.
An alternative path is:

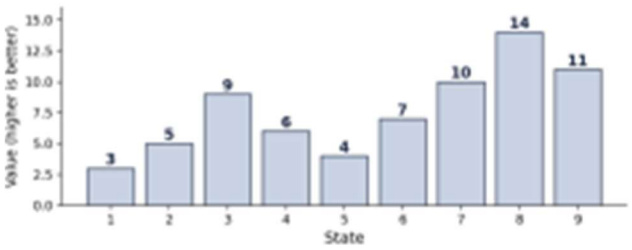
$$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$$

Its total cost is:

$$2 + 2 + 3 + 1 = \boxed{8}$$

Since 8 is less than 11, the alternative path is cheaper

Question 10 – Hill Climbing (Steepest Ascent)



Task: The chart shows a one-dimensional state space: each state's neighbours are the states on its left and right. Run steepest-ascent hill climbing (maximise value) (a) from state 2 and (b) from state 6.

Run	Current State (Value)	Neighbours (Values)	Best Neighbour	Move or Stop
1	2 (5)	1 (3), 3 (9)	3 (9)	Move to 3
2	3 (9)	2 (5), 4 (6)	None	Stop
3	6 (7)	5 (4), 7 (10)	7 (10)	Move to 7
4	7 (10)	6 (7), 8 (14)	8 (14)	Move to 8
5	8(14)	7(10),9(11)	None	Stop

Questions:

1. Where does each run stop? Is it a local or a global maximum?
- Run (a): State 3, value 9 — local maximum.
- Run (b): State 8, value 14 — global maximum.

2. How can random-restart hill climbing reach the global maximum?
Random-restart hill climbing starts the search from different randomly selected states. If one run becomes trapped at a local maximum, another run may reach the global maximum

Question 13 – Genetic Algorithm

Task: A genetic algorithm searches for the string 111111. Fitness = number of 1s. The population is P1 = 111001, P2 = 101100, P3 = 100100, P4 = 000010. Use roulette-wheel selection with random numbers 0.15, 0.62, 0.35, 0.85. Pair the selected parents 1st with 2nd (crossover after gene 3) and 3rd with 4th (crossover after gene 2). Mutate gene 6 of the first child.

Individual	String	Fitness	Selection Probability	Cumulative Probability
P1	111001	4	4/10 = 0.40	0.40
P2	101100	3	3/10 = 0.30	0.70
P3	100100	2	2/10 = 0.20	0.90
P4	000010	1	1/10 = 0.10	1.00

Questions:

1. Which parents are selected for the four random numbers?

Using roulette-wheel selection:

- Random number 0.15 falls in P1's interval (0 to 0.40).
- Random number 0.62 falls in P2's interval (0.40 to 0.70).
- Random number 0.35 falls in P1's interval (0 to 0.40).
- Random number 0.85 falls in P3's interval (0.70 to 0.90).

Selected parents: P1, P2, P1, P3.

The pairs are:

- First pair: P1 and P2.
- Second pair: P1 and P3.

2. Write the four children after crossover.

First pair: Crossover after gene 3
P1 = 111|001
P2 = 101|100
Exchange the parts after gene 3.

- Child 1 = 111100
- Child 2 = 101001

Second pair: Crossover after gene 2
P1 = 11|1001
P3 = 10|0100
Exchange the parts after gene 2.

- Child 3 = 110100
- Child 4 = 101001

Four children after crossover:

Child	String

Child 1	111100
Child 2	101001
Child 3	110100
Child 4	101001

3. Write the first child after mutation.
First child before mutation:
111100
Gene 6 is 0. Mutation flips it to 1.
First child after mutation:
111101
The final population is:

Child	Final String	Fitness
Child 1	111101	5
Child 2	101001	3
Child 3	110100	3
Child 4	101001	3
Total		14

4. Compare the average fitness before and after one generation.
Before:

$$\frac{4+3+2+1}{4} = \frac{10}{4} = \boxed{2.5}$$

After crossover and mutation:

$$\frac{5+3+3+3}{4} = \frac{14}{4} = \boxed{3.5}$$

Answer: Average fitness increases from 2.5 to 3.5.

Question 20 – Alpha–Beta Pruning

Task: Apply alpha–beta pruning to the same tree, from left to right. Start with $\alpha = -\infty$ and $\beta = +\infty$ at A. Record α and β at each node and mark every pruned leaf.

Step	Node	α (Alpha)	β (Beta)	What Happens
1	A (MAX)	−∞	+∞	Start at the root
2	B (MIN)	−∞	+∞	Explore the left subtree
3	D (MAX)	−∞	+∞	Evaluate child nodes
4	E (MAX)	Updated value	Updated value	Update α and β
5	C (MIN)	Updated value	Updated value	Explore the right subtree
6	F (MAX)	Updated value	Updated value	Evaluate child nodes
7	G (MAX)	Updated value	Updated value	Check the pruning condition
8	A (MAX)	Final α	Final β	Determine the final minimax value

Questions:

1. Which leaves are pruned?
The exact leaf labels require the missing tree and utility values. A branch is pruned whenever $\alpha \geq \beta$.

2. How many leaves are examined?
The exact count depends on the tree's leaf values and the resulting pruning decisions.

3. Is the root value the same as minimax? Why?

Yes. Alpha–Beta Pruning produces the same root value as Minimax because it removes only branches that cannot change the final decision.

It reduces the number of nodes evaluated without changing the optimal result.